BIM308 – Web Server Programming – 3rd Assignment

In this assignment, you will develop a **frontend application** using either **Angular or React** to interact with a library system. Instead of building the backend with EFCore you did in the previous assignment, you will now consume data from **JSON files.** The goal of this assignment is to understand:

- Component-based architecture (Angular / React)
- State management
- API consumption (fetch / axios / HttpClient)
- Handling relational data on the frontend
- CRUD operations in frontend applications
- UI design and user interaction

1. Project Setup

Create a new project using one of the following:

- **Angular**: ng new book-library-app-angular
- **React**: npm create-react-app book-library-app-react

Run the application to ensure it works properly.

2. Data Models

You will use the following entities:

Author:

- authorID (int)
- authorName (string)
- authorInfo (string)

User:

- userID (int)
- name (string)
- email (string)
- rentedBookIDs (array of int)

Book:

- bookID (int)
- title (string)
- releaseYear (int)
- price (number)
- authorID (int)

3. Data Source

Use these JSON files for the project:

- Authors: https://raw.githubusercontent.com/emindk/JSON_Files/refs/heads/main/authors.json?token=GHSAT0AAAAAAD2ANKTXWULL2ONOI4WIINC62OY3DKA

- Books: https://raw.githubusercontent.com/emindk/JSON_Files/refs/heads/main/books.json?token=GHSAT0AAAAAAD2ANKTWCZJORGRY2CXGKFS22OY3KFQ
- Users: https://raw.githubusercontent.com/emindk/JSON_Files/refs/heads/main/users.json?token=GHSAT0AAAAAAD2ANKTWSOGO7AVNGLPY7LWS2OY4D4A

4. Application Features

4.1 Book List Page (Index): Display all books with their title, author name (**not ID**), price and image

4.2 User List Page (Users): Display all users with their name, email and rented book's names (**not bookID**).

4.3 Rent Book Feature: Allow a user to rent a book. Update user's rentedBookIDs when finished.

Submitting the Assignment

Zip up your project which should be named as **yourlastname_yourID_hw3.zip** and submit it to ESTUOYS platform until April 19 at 23:59.

Only one group member should make submission. Also, please provide a **ReadMe.txt** file that gives information about the group members. Otherwise, your colleague might not receive any points.

Do not forget to put your Project into a container by using the **Dockerfile** that is given to you. Be careful! You will need to modify your Dockerfile and change the last line from "**BookLibrary.dll**" to "**BookLibraryReactAngular.dll**".

Late submission is not allowed by the system. Submissions via e-mail will not be accepted by the instructors and they will be deleted immediately.